

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Technology Stack

Define Your Technology Stack:
Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.
Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	ServiceNow Next Experience UI / Classic UI, AngularJS, HTML5, Bootstrap	Provides dynamic form rendering, responsive list interfaces, real-time client validation, and conditional UI action visibility (e.g., 'New' and 'Delete' buttons).
2	Backend / Server-Side	ServiceNow Server-Side JavaScript (ECMAScript 2021 / ES512 mode), GlideSystem (gs) API	Executes secure, server-side access evaluations using gs.hasRole() and GlideRecord context to enforce row-level and field-level security before query delivery.
3	Database / Data Storage	ServiceNow Relational Database Architecture (MariaDB / Enterprise Relational DB backend)	Underpins custom table schemas (u_institution_details) and system authorization tables (sys_user, sys_user_role, sys_security_acl) with high ACID compliance.
4	Cloud / Hosting / Deployment	ServiceNow Multi-Instance Cloud Platform (PaaS / SaaS Enterprise Cloud)	Offers high availability, enterprise-grade isolated data infrastructure, automated rollbacks, and scalable cloud performance.
5	Version Control & CI/CD	ServiceNow Application Scope & Update Sets ('ACL Project [Global]') / Studio	Enables configuration tracking, auditability, artifact packaging, and controlled movement of ACL rules and custom roles between Dev, Test, and Prod environments.
6	Security & Admin Tools	Elevated Privilege Framework (security_admin), Impersonation Engine, ACL Debugger	Empowers security administrators to safely modify high-security settings, test role-based access perspectives in real time, and audit permission evaluation traces.